

TrajWAM: An Efficient Trajectory-Guided World Action Model for Dynamic Robotic Manipulation

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Abstract—Dynamic object manipulation requires robots to anticipate target motion while replanning fast enough to avoid executing outdated actions. Existing approaches either rely on fast reactive control without an explicit future target state or encode anticipated motion only implicitly. World Action Models (WAMs) offer a promising alternative by coupling future visual prediction with action generation, but scene-level video objectives may underrepresent target dynamics that are critical for manipulation, while iterative video-action denoising introduces substantial latency. We present TrajWAM, a trajectory-guided world action model for efficient dynamic robotic manipulation. Our key insight is to explicitly ground future visual imagination in predicted target-object motion and use the resulting future representation to condition action generation, while preserving this anticipation pathway after acceleration. Given a short history of object masks, a history-conditioned motion predictor forecasts future masks encoding the target object’s trajectory. A lightweight Trajectory ControlNet then steers a pretrained video model toward trajectory-consistent futures. The resulting future representation conditions an action expert to predict executable action chunks. To enable one-step inference without breaking this interface, we introduce a decoupled distillation scheme that combines distribution matching for the video backbone, Trajectory ControlNet realignment, and action rollout distillation to support accurate one-step action prediction conditioned on the compressed future representation. Comprehensive evaluations in simulation and the real world show that TrajWAM achieves state-of-the-art performance on dynamic object manipulation tasks while enabling efficient one-step video-action inference.

I. INTRODUCTION

Dynamic object manipulation is essential for robots operating in human environments, factories, and warehouses, where task-relevant objects may continue moving while the robot perceives and acts [1]. Unlike static manipulation, in which the target state is approximately fixed between observation and execution, dynamic manipulation couples perception, prediction, and control in a continuously changing scene. An action that is appropriate for the current observation may therefore become outdated before execution, and even a small delay may lead to a missed grasp or inaccurate contact. Successful dynamic manipulation consequently requires both anticipation of the target object’s motion and sufficiently fast closed-loop replanning.

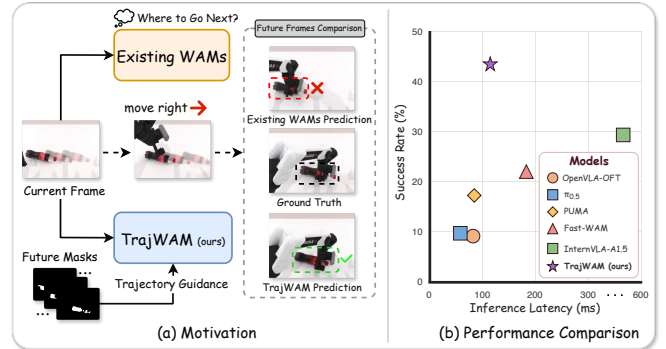


Fig. 1. Motivation and performance comparison. (a) Existing WAMs may fail to accurately anticipate target-object motion, resulting in future predictions that deviate from the actual trajectory. TrajWAM uses predicted future object masks as trajectory guidance to generate motion-consistent futures for action prediction. (b) Success rate and inference latency on DOMINO. TrajWAM achieves state-of-the-art performance while maintaining efficient inference.

World Action Models (WAMs) have recently emerged as a promising alternative to vision-language-action policies by coupling future visual prediction with action generation [2], [3], [4]. By modeling how observations evolve under interaction, WAMs provide an additional physical prior for control and are particularly appealing for dynamic manipulation. However, as illustrated in Fig. 1, existing WAMs predict scene evolution at the full-frame level, where robot-induced changes can dominate the imagined future. In dynamic manipulation, task success may instead depend critically on target motion that evolves independently of the robot’s actions. A scene-level forecast entangles robot motion, background dynamics, and target-object motion, making the task-relevant trajectory only indirectly represented and difficult to control explicitly during generation. Extending WAMs to dynamic object manipulation thus calls for an object-centric future representation that specifies the anticipated spatial trajectory of the target while remaining coupled to action prediction.

Moreover, most existing WAMs synthesize future videos and actions through iterative denoising, introducing latency that can make the resulting prediction stale before execution. Recent methods reduce this cost either by skipping future imagination during inference [5], [6] or by compressing the denoising process [7], [8], [9]. However, the former sacrifices an explicit predictive pathway for anticipating target motion, while the latter leaves open whether a spatially explicit motion condition can remain effective after aggressive denoising compression.

To address these challenges, we propose **TrajWAM**, a

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trajectory-guided world action model for efficient dynamic robotic manipulation. Our key insight is to explicitly incorporate predicted target-object motion into future visual imagination and propagate its influence to action generation through the resulting video representation. Starting from a short observation history, TrajWAM predicts how the task-relevant object will evolve and represents this motion as a sequence of future object masks. These predicted masks provide an explicit spatial guide for future generation, steering the world model toward futures that are consistent with the anticipated target motion. The resulting imagined future then conditions action prediction, allowing the robot to act not only on what is currently observed, but also on where the target is expected to be.

The remaining challenge is to make this anticipation pathway fast enough for dynamic control. Simply bypassing future imagination would remove the mechanism through which predicted target motion influences the action, while naively compressing the generative process can degrade both trajectory controllability and downstream action prediction. We therefore introduce a decoupled one-step distillation strategy that accelerates the video and action generation processes according to their different requirements. The video pathway is distilled while explicitly preserving its response to trajectory guidance, whereas the action pathway is distilled to retain accurate predictions under the compressed future representation. In this way, TrajWAM substantially reduces inference latency without discarding the trajectory-guided future imagination that enables anticipatory control.

We evaluate TrajWAM on dynamic manipulation tasks in both simulated and real-world settings, covering diverse object categories, motion patterns, and task types. On DOMINO, as shown in Fig. 1(b), TrajWAM achieves state-of-the-art performance with a success rate of 43.54%, outperforming the strongest evaluated baseline by 14.24 percentage points at an inference latency of 108.5 ms. In real-world experiments, TrajWAM achieves success rates of 77.9% on conveyor-belt grasping and 70.4% on rotating-disc grasping, consistently outperforming the evaluated baselines.

Our contributions are summarized as follows:

- We introduce TrajWAM, a trajectory-guided world action model that explicitly incorporates predicted target-object motion into future visual imagination and propagates this anticipation to downstream action prediction.
- We develop a decoupled one-step distillation strategy that accelerates video and action generation while preserving trajectory controllability and accurate future-conditioned action prediction.
- Extensive experiments in simulation and on real robots establish state-of-the-art dynamic manipulation performance while demonstrating efficient one-step inference and the effectiveness of trajectory-guided anticipation.

II. RELATED WORK

a) World Action Models and Dynamic Robotic Manipulation: Vision-language-action models have made substantial progress in robot manipulation by directly mapping visual

observations and language instructions to actions [10], [11]. World-action models (WAMs) extend this paradigm by jointly modeling future observations and executable actions, using visual dynamics prediction as an additional physical prior for control. Recent WAMs have studied different ways to balance future imagination and action execution. Fast-WAM [5] shows that explicit future generation can be skipped at inference, while AHA-WAM [9] and HarmoWAM [12] further improve the temporal coordination between world prediction and action generation. Dynamic manipulation requires anticipating the motion of objects that move independently of the robot. DOMINO [13] exposes the resulting performance degradation, while DynamicVLA [14] and PUMA [13] address it through low-latency inference and implicit object-motion prediction, respectively. DynamicWAM [15] incorporates recent motion into WAMs through dual-path flow conditioning, whereas TrajWAM predicts future target-object mask trajectories to guide future imagination for action prediction.

b) Trajectory-Controllable Video Generation:

Trajectory-controllable video generation aims to synthesize videos that follow explicit spatial motion conditions, complementing text- or image-conditioned generation where object motion is often under-specified. Existing methods use point tracks or trajectories for fine-grained motion control [16], bounding boxes or tracklets to guide object locations [17] and interactions [18], and 3D or depth-aware trajectories to reduce ambiguity in object motion [19]. More recently, MagicMotion [20] proposes dense-to-sparse trajectory guidance with masks, boxes, and sparse boxes, improving trajectory adherence and object consistency. We build on this idea of explicit motion conditioning, but use it as a future-imagination mechanism inside WAMs: future object masks are predicted from observation history and injected into the video branch to support downstream dynamic manipulation.

c) Distillation for Diffusion-Based Models: Dynamic manipulation requires low-latency inference, since the policy must continuously update actions while objects move. This is challenging for diffusion-based WAMs, where both future imagination and action generation rely on iterative denoising. Diffusion distillation provides a common way to reduce sampling steps [21], [22], [23], and has been extended to efficient video generation [24], [25] and trajectory-controllable generation [26]. Recent WAMs also explore this direction: Flash-WAM [8] proposes modality-aware step distillation for joint video-action diffusion, while AHA-WAM-Flash [9] applies ODE distillation to compress its action sampler from multi-step denoising to few-step inference. While these works primarily improve the efficiency of existing WAM inference pipelines, preserving trajectory controllability and action accuracy under compression remains underexplored.

III. METHOD

TrajWAM connects anticipated target-object motion to action generation through controlled video generation. As shown in Fig. 2, given a language instruction and a short observation history, TrajWAM first obtains the mask history

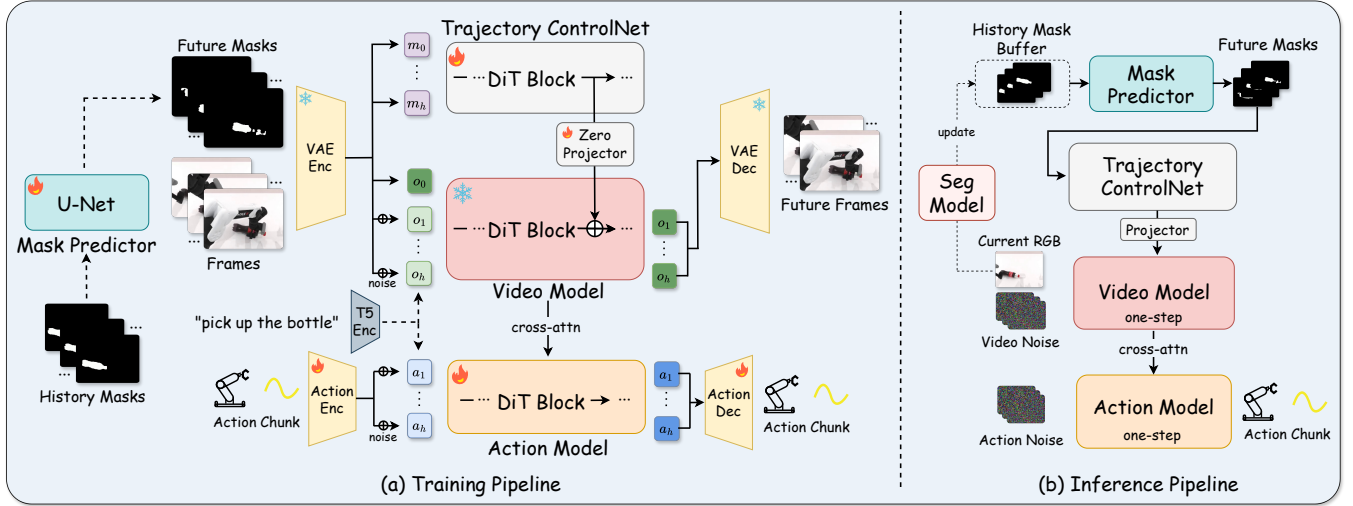


Fig. 2. Overview of TrajWAM. (a) **Training pipeline.** A language-conditioned segmentation model extracts the mask history of the target object, from which a motion predictor forecasts a sequence of future masks representing its anticipated trajectory. The predicted mask trajectory guides future-video generation through a Trajectory ControlNet, and the resulting trajectory-controlled video representation conditions the action branch through asymmetric video-action attention to predict an executable action chunk. (b) **Inference pipeline.** Target-object masks are updated online from incoming observations and maintained in a short history buffer. The predicted future mask trajectory guides the distilled one-step video model, whose future representation conditions the one-step action model for low-latency closed-loop control.

of the referred object and forecasts a sequence of future masks representing its anticipated trajectory. The predicted mask trajectory then guides future video generation, and the resulting trajectory-controlled video representation conditions the action expert to predict executable robot actions. However, the resulting world-action model remains too slow for dynamic manipulation, where excessive inference latency can make the predicted target state and corresponding action outdated before execution. We therefore distill TrajWAM into a one-step model to enable sufficiently fast closed-loop control for manipulating moving objects.

A. Object-Centric Future Trajectory Prediction

To provide the video model with object-centric dynamics, we use a frozen pretrained language-conditioned segmentation model $\mathcal{S}$ to extract the target-object mask $m_i = \mathcal{S}(o_i^{3rd}, \ell)$ from the third-person camera's view, where ℓ denotes the language instruction. We use the fixed third-person view because it provides a stable global reference for observing the target-object trajectory. In contrast, the wrist-mounted cameras move with the robot arms and may intermittently lose sight of the target, making them unsuitable for maintaining a consistent mask history. The wrist views therefore provide RGB observations only and do not provide mask inputs.

Given a causal history M_t^- , which consists of K masks sampled with temporal stride s up to the current time t , a compact 2D U-Net [27] g_ϕ predicts all H_v future masks in one forward pass:

$$M_t^+ = \sigma(g_\phi(M_t^-)), \quad (1)$$

where σ denotes the element-wise sigmoid and M_t^+ contains foreground probabilities. The predictor is trained with binary cross-entropy on the logits and soft Dice loss [28] on the predicted probabilities, each weighted by 0.5, to balance

pixel-wise foreground classification with region-level mask overlap.

B. Trajectory-Conditioned Video-Action Modeling

The predicted mask trajectory provides explicit information about how the referred object will move. TrajWAM injects this trajectory into the video branch and lets the action branch read the resulting controlled video tokens through asymmetric joint self-attention.

a) *Trajectory ControlNet:* As shown in Fig. 2(a), the predicted mask trajectory is encoded by the pretrained video VAE into trajectory latents $z_M = \mathcal{E}_{\text{vae}}(M_t^+)$. A lightweight Trajectory ControlNet [20] maps these latents and the video flow time to layer-wise residuals. Specifically, our video model is initialized from Wan2.2-5B [30], and the Trajectory ControlNet is randomly initialized while retaining the same depth as Wan2.2-5B but reducing its hidden width from 3,072 to 512. Conditioned on z_M and the video flow time s_v , the Trajectory ControlNet injects residuals into each video block:

$$\tilde{h}_v^j = h_v^j + \mathcal{Z}_j(A_\varphi^j(z_M, s_v)), \quad (2)$$

where h_v^j is the hidden state of the j -th video block, A_φ^j is the corresponding ControlNet block, and $\mathcal{Z}_j$ is a zero-initialized convolutional projection.

b) *Asymmetric Video-Action Attention:* The action expert is initialized from the pretrained Wan2.2-5B, retaining the same depth while reducing its hidden width from 3,072 to 1,024. At each transformer block, video and action tokens are concatenated for self-attention:

$$[h_v; h_a] = \text{softmax}\left(\mathcal{M}\left(\frac{QK^\top}{\sqrt{d}}\right)\right)V, \quad (3)$$

where an asymmetric attention mask $\mathcal{M}(\cdot)$ controls the information flow: video queries attend only to video keys and values, whereas action queries attend to both action and

video keys and values. Consequently, the action branch reads the trajectory-controlled future-video representation, while the video prediction remains independent of action tokens.

c) Training Objective: During training, the video backbone, VAE, and text encoder are frozen, while the Trajectory ControlNet, zero-initialized projection layers, and action expert are updated. Let $x_v = y_{>t}$ and $x_a = a_{>t}$ denote the clean future video latents and ground-truth action chunk, respectively. For each modality $q \in \{v, a\}$, we independently sample $\epsilon_q \sim \mathcal{N}(0, I)$ and $s_q \sim p(s)$, and construct noisy inputs $x_q^{s_q} = (1 - s_q)x_q + s_q\epsilon_q$. Given these inputs, their flow times, and the condition $C = (y_t, z_M, c)$, the model predicts video and action velocities $\hat{u}_v$ and $\hat{u}_a$ with the objective

$$\mathcal{L} = \mathbb{E} \left[\sum_{q \in \{v, a\}} \lambda_q \|\hat{u}_q - (\epsilon_q - x_q)\|_2^2 \right], \quad (4)$$

where y_t and c denote the current-frame latent and language condition, and λ_v, λ_a balance action learning and video co-training. Thus, ground-truth future actions provide supervision for the action branch, while the trajectory condition influences action prediction through the controlled video prediction in the same joint forward pass.

C. One-Step Deployment via Decoupled Distillation

The trajectory-conditioned model above relies on iterative flow sampling for future video and action generation; its inference latency is difficult to reconcile with the control frequency required for closed-loop dynamic manipulation. Compressing both generation processes to a single step introduces two coupled changes: the video generator loses the refinement of multi-step sampling, and the action branch must read a different video representation. We therefore distill the model in three decoupled stages: first compress the uncontrolled video generator, then realign the Trajectory ControlNet with this one-step generator, and finally adapt the action branch to the deployed trajectory-conditioned generation process.

a) Video Backbone Distillation: We first accelerate the video model independently of the Trajectory ControlNet and action branch, as illustrated in Fig. 3(a). The one-step generator G_ψ produces future latents $\hat{y}^0 = G_\psi(\epsilon_v | y_t, c)$ from Gaussian noise ϵ_v , where y_t denotes the current frame latent and c denotes the language condition. Following distribution matching distillation (DMD) [31], we perturb student samples as $\hat{y}^\tau = (1 - \tau)\hat{y}^0 + \tau\epsilon$, with a random noise level τ and $\epsilon \sim \mathcal{N}(0, I)$. A frozen real-score model f_Θ and a trainable fake-score model f_ω yield clean predictions:

$$\begin{aligned} \hat{y}_{\text{real}}^0 &= \hat{y}^\tau - \tau f_\Theta(\hat{y}^\tau, \tau | C_{\text{vid}}), \\ \hat{y}_{\text{fake}}^0 &= \hat{y}^\tau - \tau f_\omega(\hat{y}^\tau, \tau | C_{\text{vid}}). \end{aligned} \quad (5)$$

Their normalized difference defines the DMD direction

$$\Delta_{\text{DMD}} = \frac{\hat{y}_{\text{fake}}^0 - \hat{y}_{\text{real}}^0}{\text{mean}(|\hat{y}^0 - \hat{y}_{\text{real}}^0|) + \varepsilon}, \quad (6)$$

where $\varepsilon > 0$ is a small positive constant added for numerical stability. The generator is optimized with the surrogate

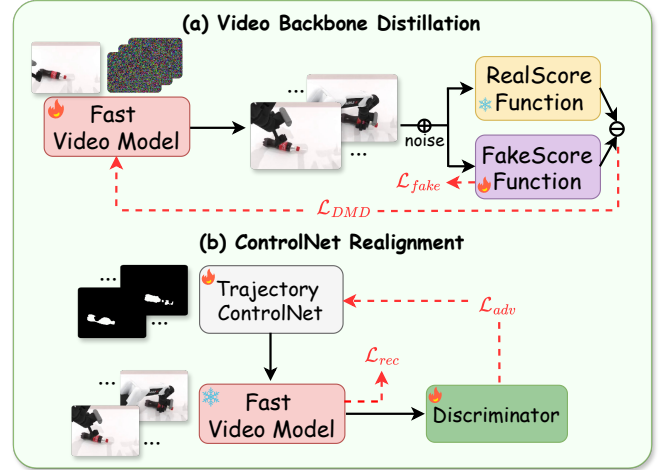


Fig. 3. Video distillation. (a) Video backbone distillation compresses the multi-step video generator into a fast generator using a frozen real-score function and a trainable fake-score function under distribution matching distillation. (b) ControlNet realignment adapts the Trajectory ControlNet to the frozen fast video model using latent reconstruction and diffusion-discriminator objectives.

objective

$$\mathcal{L}_{\text{DMD}} = \mathbb{E} \left[\frac{1}{2} \|\hat{y}^0 - \text{sg}[\hat{y}^0 - \Delta_{\text{DMD}}]\|_2^2 \right], \quad (7)$$

where sg denotes stop-gradient. The fake-score model minimizes the mean-squared error between $\hat{y}_{\text{fake}}^0$ and the detached student sample $\text{sg}[\hat{y}^0]$. We alternate generator and fake-score updates while keeping the real-score model frozen.

b) Trajectory Adapter Realignment: The slow Trajectory ControlNet was trained with the intermediate feature distribution of the multi-step video generator. After video backbone distillation, directly attaching it to the one-step generator causes a mismatch in both noise levels and intermediate activations. As illustrated in Fig. 3(b), we therefore initialize a fast trajectory adapter φ_{fast} from the trained slow Trajectory ControlNet φ_{slow} and realign it while keeping the fast video generator frozen. For a latent y^τ on the one-step schedule, the controlled generator predicts

$$\hat{y}_{\text{fast}}^0 = y^\tau - \tau u_{\psi, \varphi_{\text{fast}}}(y^\tau, \tau | y_t, z_M, c). \quad (8)$$

We train the adapter with latent reconstruction and adversarial objectives. The diffusion discriminator D_η uses a frozen pretrained multi-step video model with lightweight trainable attention modules at selected intermediate layers and a shallow MLP head. Conditioned on z_M and c , it assigns logits d_r and d_f to ground-truth and generated future latents, respectively, after perturbing both with the same Gaussian noise and noise level. The training objectives are

$$\begin{aligned} \mathcal{L}_{\text{rec}} &= \|\hat{y}_{\text{fast}}^0 - y_{>t}\|_2^2, \\ \mathcal{L}_{\text{adv}} &= \mathbb{E}[\text{softplus}(-d_f)], \\ \mathcal{L}_D &= \mathbb{E}[\text{softplus}(-d_r) + \text{softplus}(d_f)], \end{aligned} \quad (9)$$

where $y_{>t}$ is the ground-truth future video latent. The adapter is updated using $\mathcal{L}_{\text{rec}}$ and $\mathcal{L}_{\text{adv}}$, while the discriminator is updated using $\mathcal{L}_D$.

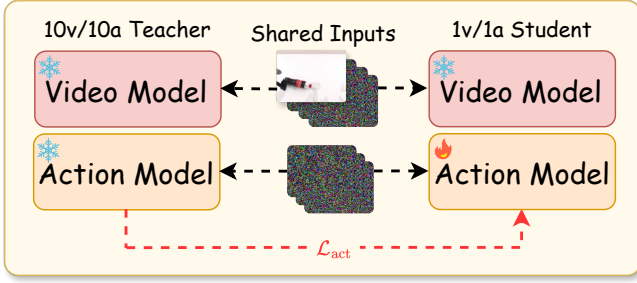


Fig. 4. Action rollout distillation. The slow teacher and fast student receive the same observation, language instruction, trajectory condition, and initial noises. The trainable fast action model is optimized to match the action chunk generated by the slow multi-step teacher.

c) *Action Rollout Distillation*: After the one-step video generator and trajectory adapter are fixed, we adapt the action branch to the deployed one-step trajectory-conditioned student, as illustrated in Fig. 4. Let $\Phi_{\theta_T}^{10,10}$ and $\Phi_{\theta_S}^{1,1}$ denote the teacher and student video-action rollouts, where the superscripts indicate the numbers of video and action denoising steps. They receive the same observation, language instruction, trajectory condition, and initial video/action noises:

$$(\hat{y}_T, \hat{a}_T) = \Phi_{\theta_T}^{10,10}(\epsilon_v, \epsilon_a \mid y_t, z_M, c), \quad (10)$$

$$(\hat{y}_S, \hat{a}_S) = \Phi_{\theta_S}^{1,1}(\epsilon_v, \epsilon_a \mid y_t, z_M, c). \quad (11)$$

The student is trained to match the action chunk produced by the teacher’s multi-step rollout, rather than an intermediate flow velocity or a ground-truth action-flow target. For a minibatch of B samples and an action chunk of T_a tokens with dimension D_a , define

$$\ell_{b,t} = \frac{1}{D_a} \sum_{d=1}^{D_a} (\hat{a}_{S,b,t,d} - \text{sg}(\hat{a}_{T,b,t,d}))^2, \quad (12)$$

where $\text{sg}(\cdot)$ stops gradients through the teacher. The training objective is

$$\mathcal{L}_{\text{act}} = \frac{1}{B} \sum_{b=1}^B \frac{\sum_{t=1}^{T_a} m_{b,t} \ell_{b,t}}{\max(1, \sum_{t=1}^{T_a} m_{b,t})}, \quad (13)$$

where $m_{b,t}$ masks padded action positions.

IV. EXPERIMENTS

A. Experimental Setup

1) *Simulation Benchmark*: We evaluate TrajWAM on DOMINO, a dynamic manipulation benchmark, which contains 35 tasks spanning different object categories, motion patterns, and manipulation objectives [13]. All methods are evaluated in a clean setting with the Aloha-AgileX robot, using a training mixture of 35 dynamic tasks under Level 1 dynamics with a dynamic coefficient of $\alpha = 0.1$. The training dataset contains 50 demonstrations per task, for a total of 1,750 demonstrations.

2) *Real-World Setup*: We further evaluate TrajWAM on two real-world dynamic grasping tasks: conveyor-belt grasping and rotating-disc grasping, as illustrated in Fig. 5. Both tasks require anticipating the target’s future motion

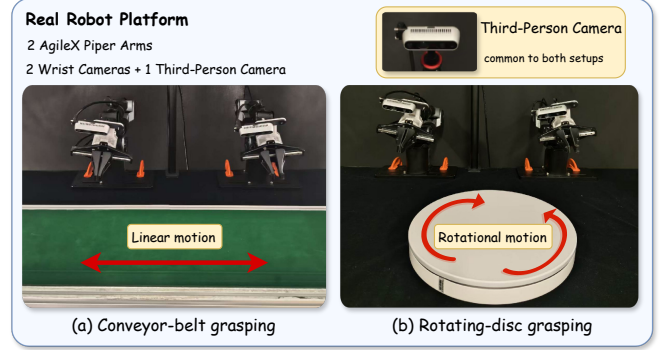


Fig. 5. Real-world evaluation setups. We evaluate TrajWAM on (a) conveyor-belt grasping with linear motion and (b) rotating-disc grasping with rotational motion. Both setups use two 6-DoF AgileX Piper arms, each equipped with a wrist-mounted Intel RealSense D435i camera, and a static D435i camera providing a global third-person view.

before the grasp is executed. The platform consists of two 6-DoF AgileX Piper arms, each equipped with a wrist-mounted Intel RealSense D435i RGB camera, and a static D435i that provides the global third-person view. Each task is evaluated at three motion speeds (low, medium, and high) using objects from four categories (banana, bottle, tape, and plush toy), covering rolling and sliding contact behaviors. We collect 200 demonstrations per task (400 in total) through teleoperation. Each demonstration consists of a single successful grasp at a randomly sampled speed using a randomly selected object. For evaluation, we run 20 trials per (task, speed, object) combination with randomized initial poses, and count a trial as successful only if the object is grasped and lifted.

3) *Implementation Details*: At inference, target-object masks are obtained using frozen SAM3 [32] and updated online during closed-loop control. In our real-world deployment, SAM3 runs asynchronously with a mean latency of 69.1 ms. At each replanning cycle, the policy uses the latest available mask history without waiting for segmentation of the current observation, while buffered actions continue to execute. The trajectory predictor takes $K = 4$ historical masks at temporal stride $s = 4$ and predicts $H_v = 8$ future masks. It is trained for 20 epochs with the BCE-Dice loss (batch size 128, initial learning rate 10^{-3}). We initialize the video model and action model from pretrained Fast-WAM-Joint [5], which uses a video backbone derived from Wan2.2-5B. During trajectory-conditioned training, we freeze the video backbone, VAE, and text encoder while optimizing the Trajectory ControlNet and action expert for 5,000 steps using a global batch size of 128 and a learning rate of 5×10^{-5} . Following Sec. III-C, we distill the video generator for 5,000 steps (global batch size 8), with generator/fake-score learning rates of $5 \times 10^{-7}/1 \times 10^{-7}$ and a 1:5 generator-to-fake-score update ratio. With the distilled video generator frozen, we realign the trajectory adapter for 1,000 steps (learning rate 2×10^{-6}), then distill the action branch for 2,000 steps (constant learning rate 10^{-6}), both with a global batch size of 4. The teacher uses 10 denoising steps per modality; the deployed student uses one. We use NVIDIA H200 GPUs for training, A800 GPUs for DOMINO evaluation, and a single NVIDIA RTX 5090 32 GB GPU for

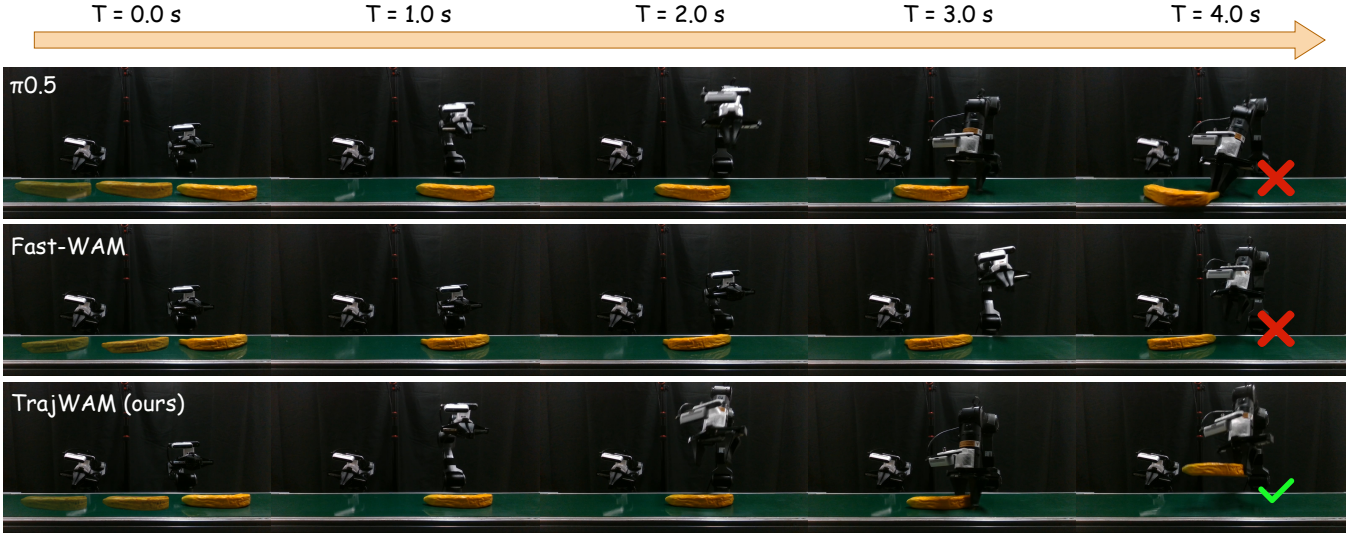


Fig. 6. Qualitative comparison of real-world conveyor-belt grasping. Snapshots are shown at 1 s intervals from $T = 0$ to 4 s. TrajWAM conditions action generation on trajectory-guided future visual representations to anticipate target motion, successfully grasping and lifting the moving banana. In the illustrated trials, $\pi_{0.5}$ and Fast-WAM fail to secure the object. Red crosses and the green check mark indicate grasp failure and success, respectively.

TABLE I
COMPARISON WITH SOTA METHODS ON DOMINO.

Method	SR (%) $\uparrow$	MS	Latency (ms)
OpenVLA [10]	1.54	6.10	173.6
π_0 -FAST [34]	3.54	20.87	119.9
VLA-Adapter [35]	4.40	24.31	83.6
RDT-1B [36]	5.34	17.71	246.3
InternVLA-M1 [37]	5.40	27.57	183.8
Isaac-GR00T [38]	6.10	28.60	76.4
π_0 [11]	8.17	23.96	52.9
OpenVLA-OFT [39]	9.06	24.06	83.1
$\pi_{0.5}$ [40]	9.63	26.17	59.1
PUMA [13]	17.20	34.97	85.4
Fast-WAM [5]	21.97	37.27	183.7
InternVLA-A1.5 [41]	<u>29.30</u>	<u>42.50</u>	552.2
TrajWAM (ours)	43.54	54.07	108.5

real-world experiments and all latency measurements. In both evaluation settings, the policy predicts $T_a = 32$ actions and executes 8 before obtaining a new observation and replanning. All real-world policies use Real-Time Chunking (RTC) [33], while DOMINO evaluation retains the synchronous protocol.

4) *Evaluation Metrics*: For simulation experiments, we report success rate (SR), manipulation score (MS) [13], and inference latency. SR measures the percentage of successful task executions. MS is a continuous metric defined by DOMINO to capture manipulation execution quality, providing a finer-grained evaluation than binary task success. For real-world experiments, we report task success rate only.

B. Main Results

1) *Simulation Results*: Table I compares TrajWAM with representative VLA and dynamic manipulation methods on DOMINO. TrajWAM achieves the best overall performance, with a success rate of 43.54% and a manipulation score of 54.07. Compared with the strongest baseline, InternVLA-A1.5, TrajWAM improves SR by 14.24 percentage points and MS by 11.57 points, demonstrating the effectiveness of explicitly modeling future target-object motion.

TrajWAM achieves these gains with an inference latency of 108.5 ms. Despite introducing explicit future trajectory prediction and controlled video generation, it is faster than Fast-WAM (183.7 ms) and substantially faster than InternVLA-A1.5 (552.2 ms), while achieving considerably higher manipulation performance. Although several VLA baselines have lower latency, they exhibit substantially lower success rates on dynamic tasks. These results indicate that TrajWAM provides a favorable trade-off between dynamic manipulation capability and inference efficiency.

2) *Real-World Results*: Table II compares TrajWAM with $\pi_{0.5}$ and Fast-WAM on two real-world dynamic grasping tasks. TrajWAM achieves success rates of 77.9% on conveyor-belt grasping and 70.4% on rotating-disc grasping, outperforming $\pi_{0.5}$ (54.2% and 44.2%) and Fast-WAM (29.6% and 27.9%). Aggregating all 480 trials per method across both tasks yields an overall success rate of 74.2% for TrajWAM, compared with 49.2% for $\pi_{0.5}$ and 28.8% for Fast-WAM. When results are pooled across the four object categories, TrajWAM also achieves the highest success rate in all six task-speed conditions.

The advantage remains pronounced under high-speed motion, where greater target displacement during action execution makes grasping more sensitive to timing and spatial alignment. TrajWAM achieves high-speed success rates of 65.0% and 70.0% on conveyor-belt and rotating-disc grasping, respectively, exceeding the strongest baseline, $\pi_{0.5}$, by 37.5 and 35.0 percentage points. These results are consistent with the benefit of conditioning action generation on trajectory-guided future representations, which provide anticipated target locations for grasp planning. Performance nevertheless varies across objects: high-speed conveyor-belt grasping remains challenging for bottles, with TrajWAM succeeding in 7 of 20 trials.

The real-world results further complement the DOMINO evaluation. Although Fast-WAM outperforms $\pi_{0.5}$ in simulation, their ranking reverses on both physical tasks, indicating

TABLE II
REAL-WORLD TASK SUCCESS RATES.

Method	Low-Speed				Medium-Speed				High-Speed				Overall	
	Banana	Bottle	Tape	Plush toy	Banana	Bottle	Tape	Plush toy	Banana	Bottle	Tape	Plush toy	Succ.	SR (%)
<i>Conveyor-belt</i>														
$\pi_{0.5}$	15/20	18/20	13/20	19/20	9/20	10/20	8/20	16/20	3/20	0/20	5/20	14/20	130/240	54.2
Fast-WAM	14/20	6/20	11/20	18/20	5/20	2/20	3/20	8/20	1/20	0/20	1/20	2/20	71/240	29.6
TrajWAM (ours)	18/20	<u>16/20</u>	15/20	20/20	16/20	13/20	18/20	19/20	11/20	7/20	16/20	18/20	187/240	77.9
<i>Rotating-disc</i>														
$\pi_{0.5}$	12/20	5/20	8/20	17/20	6/20	4/20	11/20	15/20	5/20	0/20	10/20	13/20	106/240	44.2
Fast-WAM	4/20	1/20	6/20	16/20	2/20	2/20	9/20	14/20	1/20	0/20	2/20	10/20	67/240	27.9
TrajWAM (ours)	14/20	11/20	16/20	18/20	12/20	13/20	12/20	17/20	12/20	10/20	15/20	19/20	169/240	70.4

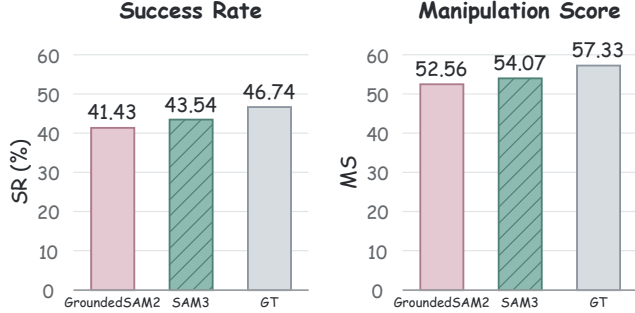


Fig. 7. Ablation of test-time mask sources on DOMINO. We evaluate success rate (left) and manipulation score (right) over 3,500 trials per source, with all model parameters fixed and only the observed masks replaced.

that baseline rankings do not necessarily carry over across evaluation settings. TrajWAM maintains its advantage in both settings, supporting its effectiveness across the evaluated dynamic manipulation tasks.

C. Ablation Studies

We conduct ablation studies on DOMINO to examine the role of trajectory conditioning and the design choices for one-step deployment, reporting SR and MS under the settings specified below.

1) *Robustness to Test-Time Mask Sources:* We evaluate the sensitivity of TrajWAM to the source of target-object masks at inference. Training uses the precomputed ground-truth masks provided with the dataset. At test time, we compare three sources for the observed mask history: simulator-provided ground-truth masks, SAM3 [32], and GroundedSAM2 [42], [43]. The trained trajectory predictor and downstream policy remain unchanged across these evaluations. For all three sources, future mask trajectories are predicted from the observed history. As shown in Fig. 7, TrajWAM maintains competitive performance across the evaluated mask sources without source-specific retraining. Replacing SAM3 with Grounded SAM 2 further decreases SR by 2.11 percentage points and MS by 1.51 points. These results suggest that TrajWAM is relatively robust to the evaluated mask sources: despite being trained with ground-truth masks, it retains most of its manipulation performance when deployed with off-the-shelf segmentation models.

2) *Effect of Test-Time Trajectory Conditions:* We compare three trajectory conditions at inference on a fixed subset

TABLE III
ABLATION STUDIES ON DOMINO.

Variant	SR (%)	MS
<i>(a) Trajectory conditions (10 tasks)</i>		
Static	20.90	33.72
Black (all-zero)	32.00	44.32
Predicted	47.30	58.32
<i>(b) One-step action distillation (12 tasks)</i>		
Undistilled (1-step inference)	7.39	15.82
DMD	1.08	19.10
CD	18.42	36.33
Action rollout (ours)	42.25	53.38
<i>(c) Trajectory adapter realignment (10 tasks)</i>		
Without realignment	42.20	52.76
With realignment	43.20	54.11

of 10 DOMINO tasks: *Predicted* uses the predicted future masks; *Static* repeats the current mask; and *Black* uses all-zero masks. All other settings are kept unchanged. As shown in Table III(a), *Predicted* outperforms *Static* and *Black* by 26.40 and 15.30 percentage points in SR, respectively, supporting the benefit of predicted motion conditions for the deployed policy. *Static* performs worse than *Black*, possibly because repeating the current mask imposes an incorrect stationary trajectory that misguides future imagination as the target moves. *Black*, although providing no explicit trajectory information, avoids this persistent misleading guidance.

3) *Comparison of Action Distillation Strategies:* We investigate how different action distillation strategies affect manipulation performance under one-step inference. All variants are evaluated on the same fixed subset of 12 DOMINO tasks, with all other settings kept unchanged. Observed mask histories are obtained from simulator-provided ground-truth masks, while future masks are still predicted by the mask predictor. We compare four variants: an undistilled action model, consistency distillation (CD) [22], distribution matching distillation (DMD) [31], and our action rollout distillation. The undistilled baseline uses the action teacher trained for 10-step sampling but directly performs inference with a single denoising step. CD and DMD train one-step action students using their respective distillation objectives. As shown in Table III(b), action rollout distillation achieves the highest SR and MS, exceeding the strongest alternative CD by 23.83 percentage points and 17.05 points, respectively. The

undistilled baseline achieves only 7.39% SR, indicating that directly applying the multi-step-trained action model with a single denoising step is insufficient for effective manipulation in this setting. DMD achieves only 1.08% SR; one possible explanation is that inaccuracies in its learned action scores compromise the distribution-matching updates, making it difficult to preserve the precise action timing and spatial alignment required for dynamic manipulation.

4) *Effect of Trajectory Adapter Realignment*: We evaluate trajectory adapter realignment on a fixed subset of 10 DOMINO tasks. The variant without realignment directly pairs the trajectory adapter trained with the multi-step video model with the distilled one-step backbone. As shown in Table III(c), realignment improves SR from 42.20% to 43.20% and MS from 52.76 to 54.11, suggesting that adapting the Trajectory ControlNet to the distilled backbone helps compensate for changes in its intermediate representations, preserving effective trajectory guidance under one-step inference.

V. CONCLUSION

We presented TrajWAM, an efficient trajectory-guided world action model for dynamic robotic manipulation. Predicted object-mask trajectories guide future video generation, providing motion-aware representations for action prediction. A decoupled distillation framework enables one-step video and action generation while preserving this conditioning pathway. TrajWAM achieves a 43.54% success rate on DOMINO with an inference latency of 108.5 ms and outperforms evaluated baselines on real-world dynamic tasks, demonstrating the effectiveness of trajectory-guided future imagination for efficient anticipatory control. However, the current experiments focus primarily on independently moving target objects; extending trajectory forecasting to action-dependent object dynamics remains an important direction toward more general dynamic manipulation.

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